Cost-Residual Prior Injection: A Prior-Agnostic Framework for Contact-Rich MPPI

Euncheol Im[1,2] Myotaeg Lim[2] and Yisoo Lee[1,†]

***Abstract*—Abstract goes here**

I. INTRODUCTION

Contact-rich manipulation—wiping, polishing, deburring, and assembly—requires the robot to actively regulate physical interaction with its environment. These dynamics are inherently non-smooth, as contact modes switch when bodies make and break contact and friction introduces discontinuities that defeat the gradient assumptions of classical model-based control, making robust force-and-motion regulation an open challenge.

Learning-based force control suffers from a sim-to-real gap, limited generalization across diverse contact dynamics, and brittleness under distribution-shifted conditions, which together make stable deployment in contact-rich environments challenging [1], [2]. Conventional model-based methods—impedance and admittance control, F/T-based disturbance modeling, and hybrid force/position control—have long been used in contact-rich tasks and provide stable interaction with the environment [3]–[5], but they ultimately rely on analytical models or estimates of the contact dynamics, which are themselves difficult to construct or estimate accurately in unstructured environments [6].

Recent work circumvents this difficulty by delegating contact behavior to a physics simulator, with a representative form being sampling-based MPC frameworks that evaluate candidate trajectories through massively parallel forward simulation [7], [8]. Among such methods, Model Predictive Path Integral (MPPI) control [9], [10] has been applied to contact-rich tasks including whole-body control of legged robots [11]. Nevertheless, the per-step cost of MPPI grows linearly with the number of rollouts K and the horizon H, so meeting the high control rates required for stable force interaction remains a bottleneck [12], [13].

To alleviate this trade-off, recent work injects learned priors into MPPI's sample-generation stage to reduce K or H required for a given task performance via normalizing flows [14], [15], diffusion models [16], RL world models [17], and flow-matching priors bootstrapped from past MPPI solutions [18], [19]. While these methods improve sample efficiency, they share a common technical pattern: each ties the prior to a specific generative architecture and biases the sampling distribution toward the prior, which can make performance sensitive to prior accuracy when the prior is out-of-distribution. We therefore seek an injection scheme that is independent of the prior's generative architecture and robust to inaccuracies in the prior, so that smaller K or H suffice for a given task performance.

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government(MSIT) (RS-2024-00339632) and by Hyundai Motor Company and Kia.

[1]E. Im and Y. Lee are with the Center for Humanoid Research, Korea Institute of Science and Technology (KIST), 02792 Seoul, South Korea

[2]E. Im and M. T. Lim are with the School of Electrical Engineering, Korea University, 02841 Seoul, South Korea

Corresponding author: Yisoo Lee yisoo.lee@kist.re.kr.

II. PRELIMINARIES: MODEL PREDICTIVE PATH INTEGRAL

MPPI [9], [10] is a sampling-based stochastic optimal control method that avoids differentiating the dynamics and cost: it evaluates many randomly perturbed control sequences in parallel and fuses them through an importance-weighted average. We review MPPI from an information-theoretic viewpoint, which makes explicit (i) the optimal *distribution* that the algorithm estimates and (ii) the quantity that governs the quality of that estimate. Both are used in the analysis of Sec. III-A.

A. Stochastic optimal control

We consider a system whose discrete-time dynamics are

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{v}_t), \tag{1}$$

where $\mathbf{x}_t \in \mathbb{R}^{n_x}$ is the state and $\mathbf{v}_t \in \mathbb{R}^{n_u}$ the applied control input. Over a horizon of T steps, the decision variable is the nominal control sequence

$$U = (\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{T-1}), \tag{2}$$

which MPPI maintains as its current estimate of the optimal control and uses as the center around which candidate inputs are sampled. Each applied control is the nominal input perturbed by zero-mean Gaussian noise,

$$\mathbf{v}_t = \mathbf{u}_t + \delta\mathbf{u}_t, \qquad \delta\mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \Sigma), \tag{3}$$

so that perturbing U yields a rollout $V = (\mathbf{v}_0, \ldots, \mathbf{v}_{T-1})$. Each rollout is scored by the trajectory cost

$$S(V) = \phi(\mathbf{x}_T) + \sum_{t=0}^{T-1} \ell(\mathbf{x}_t, \mathbf{v}_t), \tag{4}$$

with terminal cost ϕ and running cost ℓ (e.g., tracking error, control effort, or constraint penalties). The stochastic optimal control problem seeks the nominal sequence that minimizes the expected cost,

$$U^* = \arg\min_U \ \mathbb{E}_V[S(V)]\,. \tag{5}$$

B. Information-theoretic solution

Rather than seeking a single minimizer of (4), the information-theoretic formulation characterizes an optimal distribution over control sequences,

$$p^*(V) \;\propto\; \exp\bigl(-S(V)/\lambda\bigr)\, p_0(V), \tag{6}$$

where $\lambda > 0$ is the temperature and p_0 is the base distribution induced by the nominal sequence through (3), i.e., $V \sim \mathcal{N}(U, \Sigma)$. This base distribution is exactly the Gaussian distribution q from which rollouts are sampled, so $q(V) = p_0(V)$. The optimal control is the mean of this distribution, $U^* = \mathbb{E}_{p^*}[V]$. Since S is nonconvex, (6) has no closed form and is instead estimated by importance sampling. We draw K rollouts $\{V_k\}_{k=1}^K$ from q and approximate the mean by the importance-weighted average

$$\omega_k = \exp\bigl(-\tilde{S}_k/\lambda\bigr), \tag{7a}$$

$$\mathbf{u}_t^* \leftarrow \mathbf{u}_t + \frac{\sum_{k=1}^K \omega_k\, \delta\mathbf{u}_{k,t}}{\sum_{k=1}^K \omega_k}, \tag{7b}$$

where $\tilde{S}_k = S_k - \min_{j\in\{1,\dots,K\}} S_j$ shifts the costs across the K rollouts for numerical stability without altering the weight ratios. Equation (7) is precisely the self-normalized importance-sampling estimate of $\mathbb{E}_{p^*}[V]$. The normalized weight $\omega_k / \sum_{j=1}^K \omega_j$ equals $w(V_k)/\sum_{j=1}^K w(V_j)$ with $w = p^*(V)/q(V)$, and because the choice $q(V) = p_0(V)$ cancels the base-measure factor in (6), the weight reduces to a function of the trajectory cost alone, $w(V) \propto \exp(-S(V)/\lambda)$, recovering the cost-based weights in (7a).

C. Quality of the sample-based estimate

Because (7) is an importance-sampling estimate, its reliability under a finite rollout budget K depends on how well the sampling distribution q covers the optimal distribution p^*. When a few rollouts carry most of the weight, the number of effectively contributing rollouts collapses and the estimate becomes noisy. This concentration is captured exactly by the chi-squared (χ^2) divergence between target and sampling distribution [20], [21],

$$\frac{\mathrm{Var}_q[w]}{(\mathbb{E}_q[w])^2} = \chi^2\bigl(p^* \,\|\, q\bigr), \qquad w = \frac{p^*(V)}{q(V)}, \tag{8}$$

which we adopt in Sec. III-A as the measure of estimator variance: smaller χ^2 means more rollouts contribute, and $\chi^2 = 0$ if and only if $q(V) = p^*(V)$.

III. Method

A. Prior Injection as Cost Function

A learned prior provides a candidate control sequence U_p for the current state. We consider two ways of injecting U_p into the MPPI update of Sec. II. The first, used by previous work, relocates the sampling distribution so that rollouts are perturbations of the prior rather than of the nominal sequence,

$$q_\mathrm{w}(V) = \mathcal{N}(U_\mathrm{p}, \Sigma). \tag{9}$$

The second, which we adopt, leaves the baseline sampling distribution $q = \mathcal{N}(U, \Sigma)$ unchanged and instead augments the cost with a quadratic residual that penalizes deviation from the prior,

$$S'(V) = S(V) + \alpha\, \|V - U_\mathrm{p}\|^2, \tag{10}$$

where $\alpha > 0$ weights the prior. We refer to (9) as *warm-start* and to (10) as *cost-residual*.

Remark 1 (Connection to warm-start). Substituting S' into (6), the cost-residual target factorizes as

$$p'(V) \propto \exp\bigl(-S(V)/\lambda\bigr)\, \exp\bigl(-\tfrac{\alpha}{\lambda}\|V - U_\mathrm{p}\|^2\bigr)\, p_0(V), \tag{11}$$

the product of the MPPI likelihood and a Gaussian prior centered at U_p with covariance $\frac{\lambda}{2\alpha} I$. Cost-residual and warm-start are therefore two ways of injecting the same prior, one into the target and one into the sampling distribution, and the correspondence is exact in control space. The analysis below does not rely on this equivalence, since it holds in the general setting through the coverage argument.

a) Estimator variance: Under either scheme the update remains an importance-sampling estimate, and its variance is governed by the divergence $\chi^2(p\|q)$ of (8) between the effective target p and the sampling distribution q. The two schemes share the same sampling covariance Σ and differ only in the center of q, which is the prior U_p for warm-start and the nominal U for cost-residual. To make this dependence explicit, we approximate the target near its mode U^* by an isotropic Gaussian $\mathcal{N}(U^*, s^2 I)$ and take the sampling covariance to be isotropic, $\Sigma = r^2 I$, where s and r denote the target and sampling standard deviations, respectively.

Lemma 1 (Coverage divergence)**.** *For a Gaussian target $\mathcal{N}(U^*, s^2 I)$ and a sampling distribution $\mathcal{N}(\mu, r^2 I)$ with $2r^2 > s^2$,*

$$\chi^2 + 1 = \Bigl(\tfrac{r^2}{s\sqrt{2r^2 - s^2}}\Bigr)^d \exp\Bigl(\tfrac{\|\mu - U^*\|^2}{2r^2 - s^2}\Bigr). \tag{12}$$

The proof is given in Appendix A. The divergence grows exponentially with the distance from the sampling center μ to the optimum U^*, and diverges when the sampling distribution is too narrow relative to the target ($2r^2 \le s^2$).

Proposition 1 (Robustness to out-of-distribution priors)**.** *Let $\delta_\mathrm{w} = \|U_\mathrm{p} - U^*\|$ and $\delta_\mathrm{c} = \|U - U^*\|$ denote the distances from the warm-start and cost-residual sampling centers to the optimum. Under the conditions of Lemma 1 with shared $\Sigma = r^2 I$,*

$$\frac{\chi^2_\mathrm{w} + 1}{\chi^2_\mathrm{c} + 1} = \exp\Bigl(\tfrac{\delta_\mathrm{w}^2 - \delta_\mathrm{c}^2}{2r^2 - s^2}\Bigr), \tag{13}$$

so $\chi^2_\mathrm{w} > \chi^2_\mathrm{c}$ if and only if $\delta_\mathrm{w} > \delta_\mathrm{c}$.

Because the nominal sequence U is carried across control steps and refined by the update (7), it remains close to the optimum, so δ_c is small. The warm-start center U_p is fixed by the prior, so when the prior is out-of-distribution (OOD) δ_w is large. In this regime $\delta_\mathrm{w} \gg \delta_\mathrm{c}$ and the variance of warm-start grows exponentially relative to cost-residual, whereas an accurate prior ($U_\mathrm{p} \approx U^*$) makes warm-start preferable. The shared prefactor in (12) cancels in (13), so the comparison depends only on the sampling centers.

Remark 2 (Coverage interpretation). The result is independent of the space in which the prior is injected. Because cost-residual modifies only the cost and leaves the sampling distribution at the self-correcting nominal, its rollouts retain coverage of the optimum even when the prior is inaccurate, while warm-start commits all rollouts to the neighborhood of U_{p}. Cost-residual additionally avoids mapping the prior into the control space through inverse dynamics, removing a further source of error present in warm-start.

B. Framework

IV. Experiment Results

V. Conclusions

Conclusion goes here.

Appendix

We evaluate $\chi^2 + 1 = \int \big(p^*(V)\big)^2 / q(V)\, dV$, where

$$\begin{aligned} p^*(V) &= (2\pi s^2)^{-d/2} e^{-\|V-U^*\|^2/2s^2}, \\ q(V) &= (2\pi r^2)^{-d/2} e^{-\|V-\mu\|^2/2r^2}. \end{aligned} \tag{14}$$

Substituting and squaring gives

$$\begin{aligned} \chi^2 + 1 &= (2\pi s^2)^{-d} (2\pi r^2)^{d/2} \int e^{E(V)}\, dV, \\ E(V) &= -a\|V-U^*\|^2 + b\|V-\mu\|^2, \end{aligned} \tag{15}$$

with $a = 1/s^2$ and $b = 1/(2r^2)$.

Expanding and collecting the terms in V, with $A = a - b$ and $v = aU^* - b\mu$,

$$E(V) = -A\|V\|^2 + 2\langle V, v\rangle - a\|U^*\|^2 + b\|\mu\|^2. \tag{16}$$

The coefficient of $\|V\|^2$ is $-A$, so the integral converges only when $A > 0$, that is $2r^2 > s^2$. Completing the square gives

$$E(V) = -A\Big\|V - \tfrac{v}{A}\Big\|^2 + K, \qquad K = \frac{\|v\|^2}{A} - a\|U^*\|^2 + b\|\mu\|^2. \tag{17}$$

Substituting $v = aU^* - b\mu$ and $A = a - b$ and collecting terms, the self-terms $\|U^*\|^2$ and $\|\mu\|^2$ cancel and only the cross term survives,

$$K = \frac{ab}{a-b}\,\|\mu - U^*\|^2 = \frac{\|\mu - U^*\|^2}{2r^2 - s^2}, \tag{18}$$

where we used $ab/(a-b) = 1/(2r^2 - s^2)$. The remaining Gaussian integral is $\int e^{-A\|V - v/A\|^2}\, dV = (\pi/A)^{d/2}$. Collecting the constants, the factors of 2π cancel and we obtain

$$\chi^2 + 1 = \Big(\frac{r^2}{s\sqrt{2r^2 - s^2}}\Big)^d \exp\Big(\frac{\|\mu - U^*\|^2}{2r^2 - s^2}\Big), \tag{19}$$

which proves the claim. □

Acknowledgment

Conclusion goes here.

References

[1] T. Tsuji, Y. Kato, G. Solak, H. Zhang, T. Petrič, F. Nori, and A. Ajoudani, "A survey on imitation learning for contact-rich tasks in robotics," *The International Journal of Robotics Research*, p. 02783649261417694, 2025.

[2] Y. Yang, A. Chen, Z. Zhu, K. Xu, Y. Mao, Y. Wei, L. Chen, R. Xiong, and Y. Wang, "Direction matters: Learning force direction enables sim-to-real contact-rich manipulation," *arXiv preprint arXiv:2602.14174*, 2026.

[3] N. Hogan, "Impedance control: An approach to manipulation: Part ii—implementation," 1985.

[4] M. H. Raibert and J. J. Craig, "Hybrid position/force control of manipulators," 1981.

[5] L. Villani and J. De Schutter, "Force control," in *Springer handbook of robotics*. Springer, 2016, pp. 195–220.

[6] M. Suomalainen, Y. Karayiannidis, and V. Kyrki, "A survey of robot manipulation in contact," *Robotics and Autonomous Systems*, vol. 156, p. 104224, 2022.

[7] T. Howell, N. Gileadi, S. Tunyasuvunakool, K. Zakka, T. Erez, and Y. Tassa, "Predictive sampling: Real-time behaviour synthesis with mujoco," *arXiv preprint arXiv:2212.00541*, 2022.

[8] C. Pezzato, C. Salmi, E. Trevisan, M. Spahn, J. Alonso-Mora, and C. H. Corbato, "Sampling-based model predictive control leveraging parallelizable physics simulations," *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2750–2757, 2025.

[9] G. Williams, A. Aldrich, and E. A. Theodorou, "Model predictive path integral control: From theory to parallel computation," *Journal of Guidance, Control, and Dynamics*, vol. 40, no. 2, pp. 344–357, 2017.

[10] G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou, "Information-theoretic model predictive control: Theory and applications to autonomous driving," *IEEE Transactions on Robotics*, vol. 34, no. 6, pp. 1603–1622, 2018.

[11] J. Alvarez-Padilla, J. Z. Zhang, S. Kwok, J. M. Dolan, and Z. Manchester, "Real-time whole-body control of legged robots with model-predictive path integral control," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 14 721–14 727.

[12] A. Jordana, S. Kleff, J. Carpentier, N. Mansard, and L. Righetti, "Force feedback model-predictive control via online estimation," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 11 503–11 509.

[13] L. Wijayarathne, Z. Zhou, Y. Zhao, and F. L. Hammond, "Real-time deformable-contact-aware model predictive control for force-modulated manipulation," *IEEE Transactions on Robotics*, vol. 39, no. 5, pp. 3549–3566, 2023.

[14] T. Power and D. Berenson, "Learning a generalizable trajectory sampling distribution for model predictive control," *IEEE Transactions on Robotics*, vol. 40, pp. 2111–2127, 2024.

[15] J. Sacks and B. Boots, "Learning sampling distributions for model predictive control," in *Conference on Robot Learning*. PMLR, 2023, pp. 1733–1742.

[16] Y. Huang and H. Liu, "Diffusion-mppi: Diffusion informed model predictive path integral method," in *International Conference on Neural Information Processing*. Springer, 2024, pp. 284–298.

[17] P. Kotecha, S. Kolathaya *et al.*, "Real-time gait adaptation for quadrupeds using model predictive control and reinforcement learning," *arXiv preprint arXiv:2510.20706*, 2025.

[18] V. Kurtz and J. W. Burdick, "Generative predictive control: Flow matching policies for dynamic and difficult-to-demonstrate tasks," *arXiv preprint arXiv:2502.13406*, 2025.

[19] L. Brudermüller, B. Hung, X. Zhu, J. Wang, N. Hawes, P. Culbertson, and S. Le Cleac'h, "Generative models from and for sampling-based mpc: A bootstrapped approach for adaptive contact-rich manipulation," *IEEE Robotics and Automation Letters*, 2026.

[20] A. Kong, "A note on importance sampling using standardized weights," *University of Chicago, Dept. of Statistics, Tech. Rep*, vol. 348, p. 14, 1992.

[21] S. Agapiou, O. Papaspiliopoulos, D. Sanz-Alonso, and A. M. Stuart, "Importance sampling: Intrinsic dimension and computational cost," *Statistical Science*, pp. 405–431, 2017.